

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- n/a

Confirmed
- ☐

☒

The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐

☒

A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐

☒

The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐

☒

A description of all covariates tested
- ☐

☒

A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐

☒

A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐

☒

For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted
Give P values as exact values whenever suitable.
- ☒

☐

For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☐

☒

For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐

☒

Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Raw MS data were searched against the human UniProt sequence database using MaxQuant(version 1.6.1.0).
Data analysis	Differential expression analysis among various groups was tested by two-sided Student's t test. Spearman's correlation coefficients were employed to calculate the correlations between common internal references or between experimental samples. Gene Ontology (GO) term analysis and Reactome enrichment analysis were conducted via the Database for Annotation, Visualization, and Integrated Discovery (DAVID) Bioinformatics Resources. The p values for pathway enrichment analysis were calculated using the DAVID tool based on two-sided Fisher's exact test. The enrichment scores of various pathways in each sample were assessed via the ssGSEA algorithm from the GSVA package (version 1.48.3). Age- and disease-associated changes in the plasma proteome were analyzed using the DE-SWAN package (v0.0.0.9001). For treatment response prediction, LASSO regression was applied using the glmnet package (v4.1-4).Feature selection was conducted using multiple methods, including random forest (randomForest, v4.7-1.1), recursive feature elimination with SVM (caret, v7.0-1), XGBoost (xgboost, v1.7.8.1), stability selection (stabs, v0.6-4), and elastic net (glmnet, v4.1-4).To interpret model outputs, SHapley Additive exPlanations (SHAP) were calculated using the fastshap package(v1.18.5) and visualized with the shapviz package.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The raw mass spectrometry proteomics data have been deposited in the ProteomeXchange Consortium via the iProX partner repository, with the dataset identifier PXD048245 (<https://www.iprox.cn/page/DSV021.html?url=1750904980059cZK2>, Password: IHqY). All data are included in the Supplementary Information or available from the authors, as are unique reagents used in this Article. The raw numbers for charts and graphs are available in the Source Data file whenever possible. The source code, including differential computation, feature selection, prediction models and plotting, is publicly available on Zenodo at <https://doi.org/10.5281/zenodo.14851474>.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

We have provided the sex information of each participant in Source Data Fig 1 (Fig1a) (for plasma proteome) and Source Data Extended Data Fig 6 (Extended Data Fig 6g) (for ELISA validation). Sex was not a factor in our study design, as participants were enrolled randomly without prior sex-based selection or stratification. Sex of participants was determined based on self-report. We analyzed the influence of sex on protein levels as part of our study, and the corresponding source data have been provided.

Reporting on race, ethnicity, or other socially relevant groupings

In our study, no socially constructed variables were employed. All participants were recruited from hospital health check-ups or outpatient clinics, and were of Han ethnicity. Additionally, data collection relied on clinical records and assessments conducted within the hospital setting.

Population characteristics

Studies include information with sex, age, DAS28-CRP, drug, VAS, CRP, SJC and TJC. Clinical information of people is reported in Source Data Fig 1 (Fig1a) (for plasma proteome) and Source Data Extended Data Fig 6 (Extended Data Fig 6g) (for ELISA validation).

Recruitment

Patients were diagnosed with RA by meeting the 2010 American College of Rheumatology (ACR)/European League Against Rheumatism (EULAR) criteria. Individuals at risk of RA have been recruited based on an ACPA-positive response, but lacking signs and symptoms of arthritis.

Ethics oversight

These samples were collected at West China Hospital of Sichuan University, following the approval of the Research Ethics Committee of West China Hospital at Sichuan University (Permission number: 2021(790)), and informed consent was obtained from all participants and informed consent was obtained from all participants or their guardians. Human tissues for internal standards were sourced from distant normal tissues of cancer patients, with approval from the Research Ethics Committee of West China Hospital, Sichuan University (Approval numbers: 2019(538) for liver, 2019(539) for lung, and 2020 (374) for intestine). Normal kidney tissue was obtained from renal transplant donors (Approval number 2019(748)).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

Plasma samples were obtained from 99 healthy control, 60 at-risk RA, 278 patients with RA. The sample size for our proteomics experiments was determined based on data quality considerations. As shown in Figure 1d, the number of samples in each group was sufficient to achieve a protein detection rate that meets the saturation threshold of the platform.

Data exclusions

For proteomics, the proteins with at least one unique peptide were preserved. And the proteins detected in $\geq 50\%$ samples in each groups were used for subsequent analysis.

Replication

We randomly selected 10 plasma samples for technical replicates in proteomic analysis. We confirm that all attempts at replication were successful and yielded consistent results across independent experiments.

Randomization

Patients were also recruited randomly, and there was no selection bias. Our study is based on comparing rheumatoid arthritis patients, at-

risk individuals and healthy controls, and group allocation was determined by clinical status.

Blinding

For sample processing ,all investigators were blinded to the outcome.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	All antibodies used in this study were included in the commercial ELISA kits, with the capture antibodies pre-coated on the assay plates by the manufacturers. The catalog numbers of the ELISA kits were as follows: COL1A1 (Solarbio, China, Cat. No. SEKH-0401), GGT1 (Signalway Antibody, USA, Cat. No. EK14228), MHY9 (Signalway Antibody, USA, Cat. No. EK15634), RPL27A (EIAab, China, Cat. No. E5486h), ECI2 (EIAab, China, Cat. No. E16269h), LGALS3BP (Boster Biological Technology, China, Cat. No. EK1240), and CBR1 (COIBO BIO, China, Cat. No. CB16353-Hu).
Validation	All commercial antibodies used in the ELISA kits were validated by the manufacturers. Detailed information, including the source and catalog numbers of each kit, is provided in Supplementary Table 2.

Plants

Seed stocks	<i>Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.</i>
Novel plant genotypes	<i>Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.</i>
Authentication	<i>Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.</i>